SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares total site energy consumption, operational costs, construction costs, and payback period between baseline and all the applied renovation scenarios. The construction cost data is from 2026 RSMeans Database. Since RSMeans Database is proprietary, users can also adopt customized cost dataset when needed.

Building Information

- Weather File: USA_NY_Buffalo-Greater.Buffalo.Intl.AP.725280_TMY3.epw
- Building Name: SmallOffice
- Building Location: Buffalo, NY
- Total Floor Area: 511 m^2

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	**Current Condition of Exterior Wall:** • Exterior wall insulation: current R-value of insulation is R-11.3 **Current Condition of Roof:** • Roof insulation: current R-value of insulation is R-1.2 **Current Condition of Window:** • Number of window constructions is 20, with 20 simple glazing type windows, and 0 layered construction windows

		Current Condition of Door: • Number of door constructions is 3, with 1 glass door, 0 overhead door, and 2 door
Scenario 1	wall + roof + window + door	**Wall upgrade:** • Exterior wall insulation: Blown Fiberglass, target R-20.4 **Roof upgrade:** • Roof insulation: Blown Fiberglass, target R-34.5 **Window upgrade:** • 2-pane replacement • Weatherstrip application (skipped: no OperableWindow) • Glazing film application: anti-graffiti film • Caulking application: polyurethane • Window frame replacement: wood window frame **Door upgrade:** • Bottom seal: silicone adhesive smoke gasket • Top/side seal: silicone adhesive smoke gasket
Scenario 2	unknown	Envelope renovation applied (details generated at report time) **Wall upgrade:** • Requested: wall insulation upgrade to target R-20.40 using 'Blown Fiberglass'. Outcome: Implemented. Details: Exterior wall insulation layers were updated to meet the target R-value where needed. Context: modified constructions=1, target-already-met skips=0, renovated area=281.515 m2, added volume=16.988 m3. **Roof upgrade:** • Roof insulation renovation completed. Modified constructions=1, target-already-met skips=0. **Door upgrade:** • door enhancement successfully completed!

Annual Energy Analysis

Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Anual Total Site Energy (GJ)	0.00	388.03	-388.03	0.00%
Scenario 2	Anual Total Site Energy (GJ)	0.00	0.00	**0.00**	**0.00%**

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) ($/yr) Comparison

Baseline $0.00

Scenario 1 $17,052

Baseline Scenario 1

Max saving: $17,052 (0.00%)

Total Site Energy Saving (Scenario - Baseline) (GJ/yr) Comparison

Baseline 0.00

Scenario 2 0.00

Baseline Scenario 2

Max saving: **0.00 GJ (0.00%)**

Min Retrofit Construction Cost

$6,125

Scenario 1

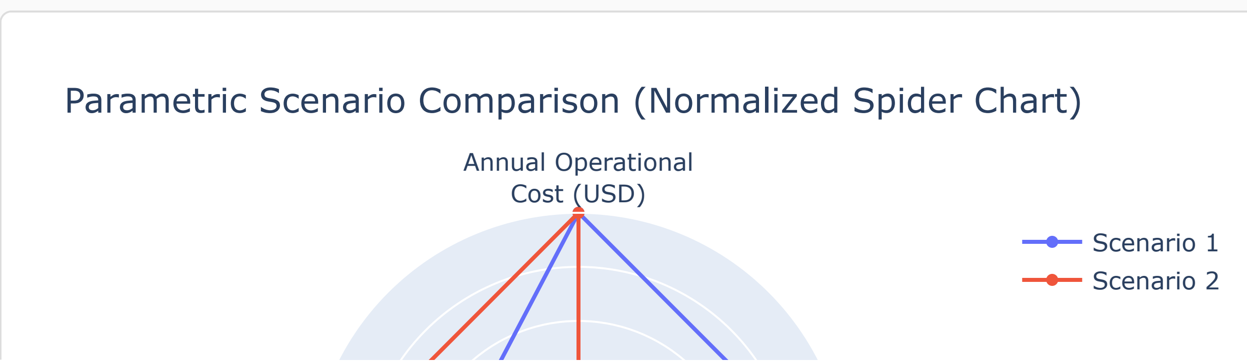
Lowest Retrofit Cost Payback Period

N/A

No valid cost payback scenario

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) ($/yr)).

Scenario 1 N/A

Scenario 2 N/A

Material List

> Material properties and consumption by renovation scenario. Missing values are shown as N/A.

Scenario	Component	Material name	Volume (m^3)	Area (m^2)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m^3)	Product Lifetime (years)
Scenario 1	Wall insulation	Blown Fiberglass	17	282	N/A	N/A	0.03	30	30
Scenario 1	Roof insulation	Blown Fiberglass	144	599	0.24	N/A	0.03	30	30
Scenario 1	Door bottom sealing	silicone adhesive smoke gasket	N/A	N/A	N/A	3.66	N/A	N/A	15

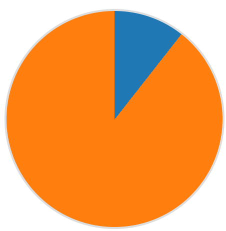
Scenario 1	Door top/side sealing	silicone adhesive smoke gasket	N/A	N/A	N/A	16	N/A	N/A	15

Result Summary

Per-scenario breakdown: left chart shows retrofit construction cost contribution by measure, right chart shows total site energy by fuel type (electricity, natural gas, and other). Water is excluded from energy breakdown.

Scenario 1

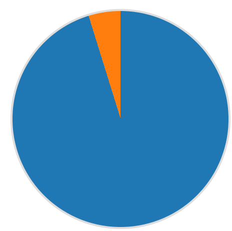
Cost Breakdown by Retrofit Measure



- Wall: $648 (10.6%)
- Roof: $5,477 (89.4%)

Total: $6,125

Annual Total Site Energy Breakdown (GJ)

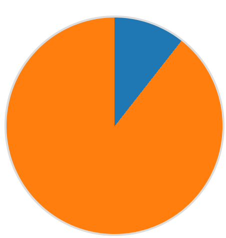


- Electricity: 369.41 GJ (95.2%)
- Natural Gas: 18.62 GJ (4.8%)

Total: 388.03 GJ

Scenario 2

Cost Breakdown by Retrofit Measure



- Wall: $1,230 (10.6%)
- Roof: $10,401 (89.4%)

Total: $11,631

Annual Total Site Energy Breakdown (GJ)

No retrofit measure contribution data.

Scenario	Retrofit Construction Cost (USD)	Annual Operational Cost (USD)	Anual Total Site Energy (GJ)
Baseline	$0.00	$0.00	0.00
Scenario 1	$6,125	$17,052	388.03
Scenario 2	$11,631	$17,052	0.00

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